

Lab 6 Report

Introduction

Beam buckling is important concept to recognize when encountering structures under load. Understanding how the structure will behave under critical conditions will help better design components. This lab was an exercise in applying buckling concepts to ANSYS Mechanical simulation. The first problem analyzes a fixed-roller beam experiencing an axial compressive load. Using a unit load, the buckling deformation and critical load were found. The second problem consisted of a truss supported by a pin and roller. Two situations were presented: a vertical load at the top point, and a horizontal load at the roller. These situations produced different buckling deformation and the analysis can be seen below. Finally, a structure consisting of a skin and stiffener/flange beams was analyzed using a vertical unit point load. All simulations were performed with Eigenvalue Buckling analysis through ANSYS Mechanical.

Data

Part 1

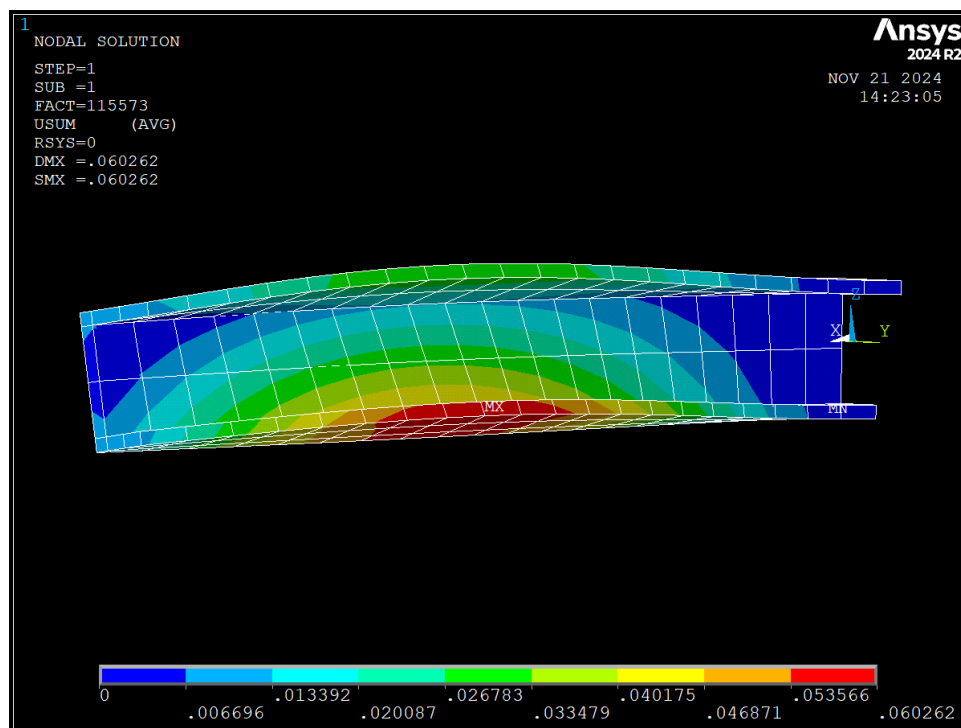


Figure 1. First Set Solution for Fixed-Roller Beam

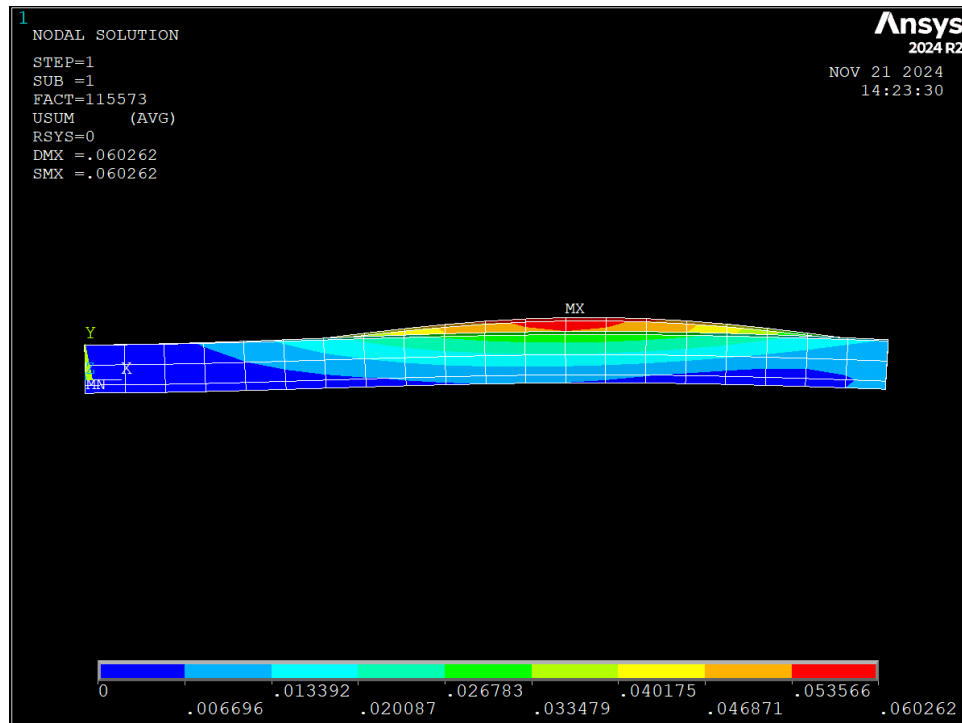


Figure 2. Top View of Node 1 Solution

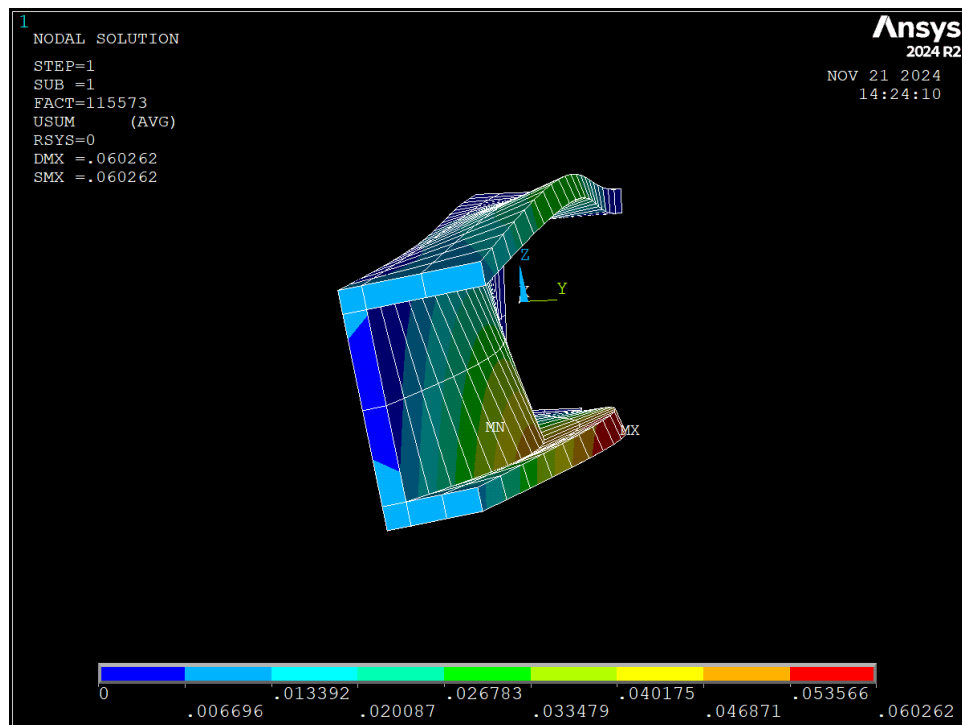


Figure 3. Front View of Node 1 Solution

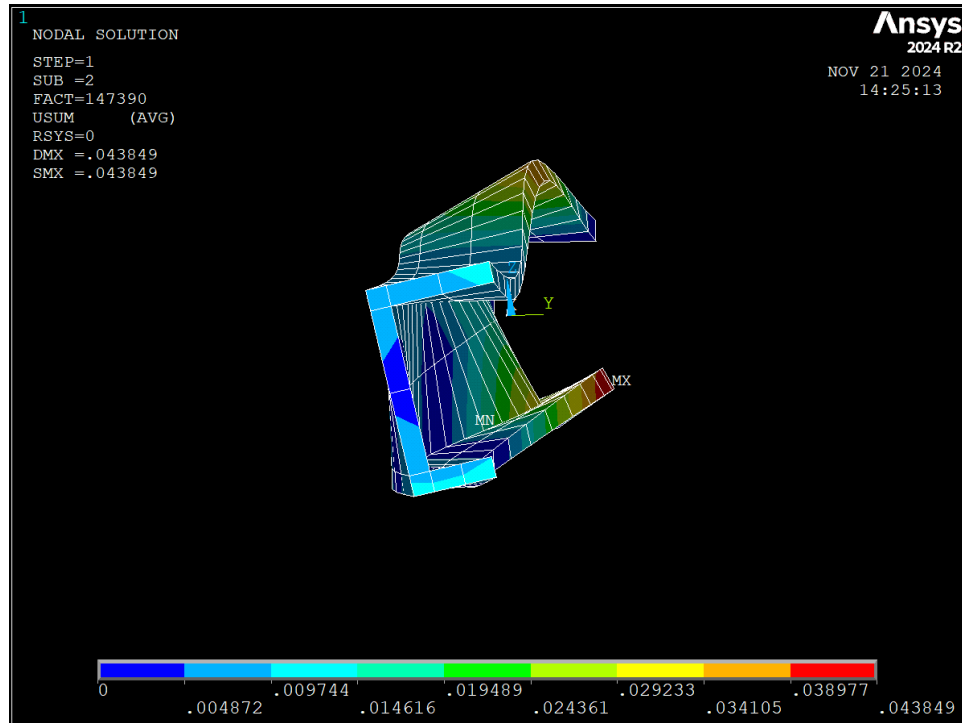


Figure 4. Front View of Node 2 Solution

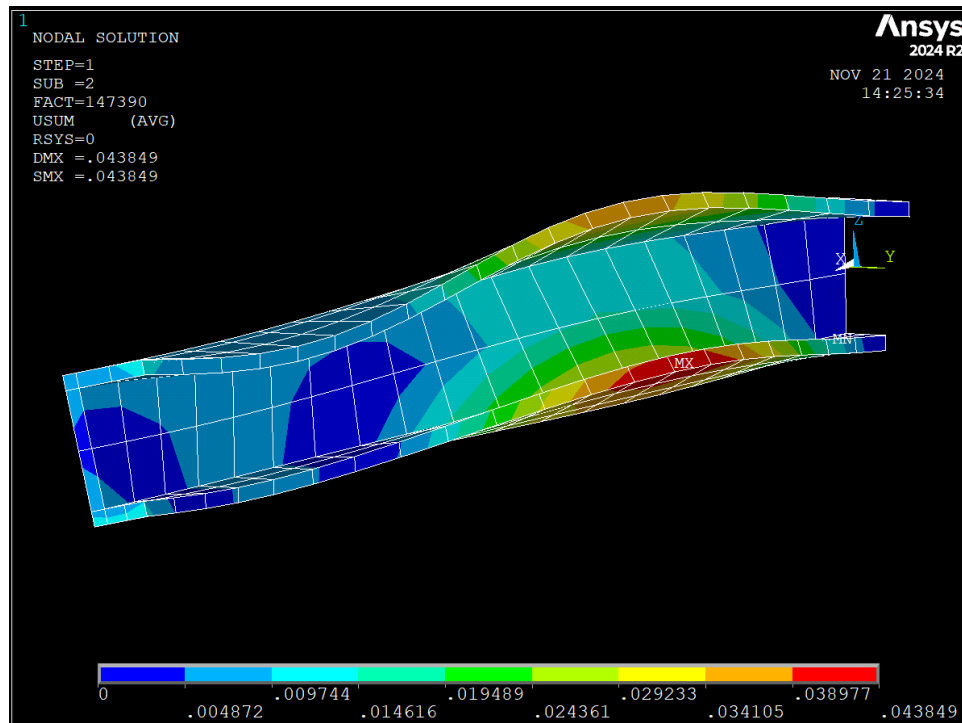
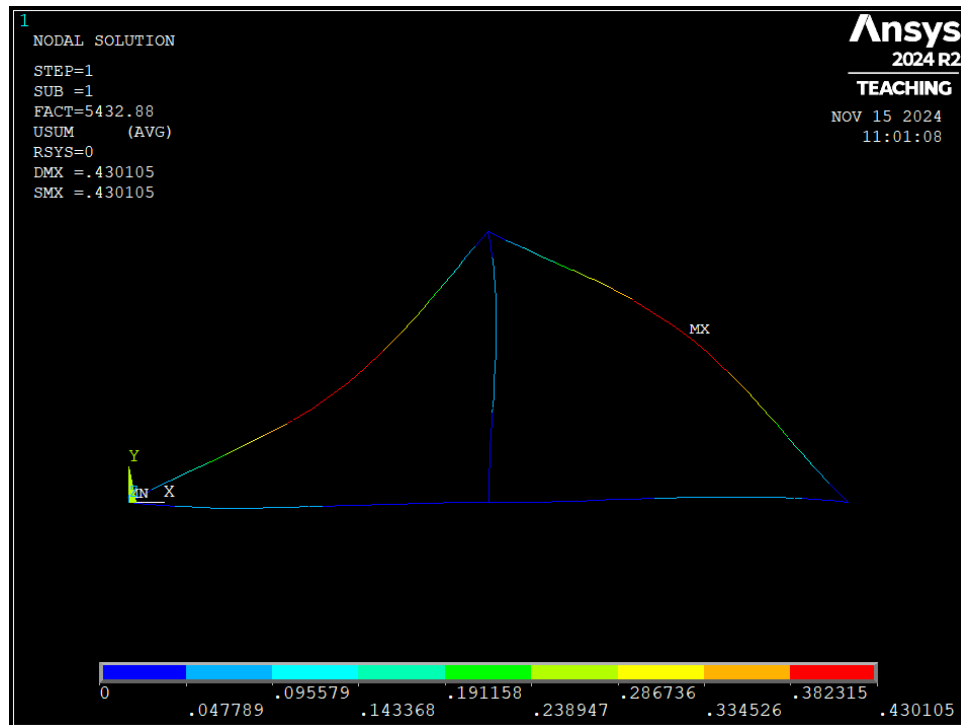
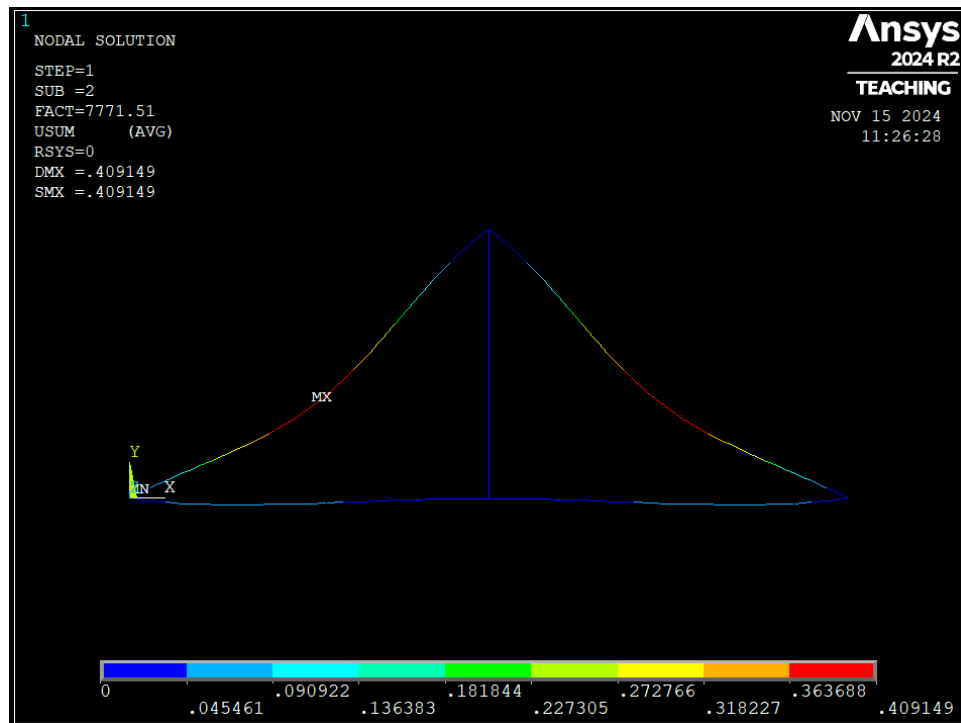


Figure 5. Side View of Node 2 Solution.

Part 2**Figure 6. Problem 1 – Mode 1 Displacement****Figure 7. Problem 1 – Mode 2 Displacement**

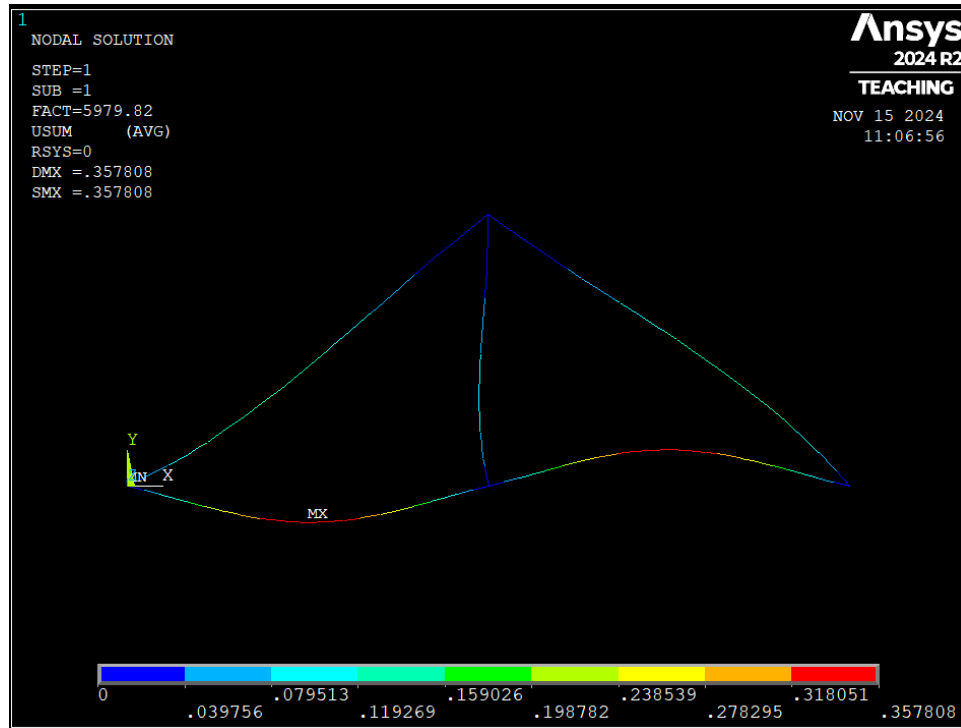


Figure 8. Problem 2 – Mode 1 Displacement

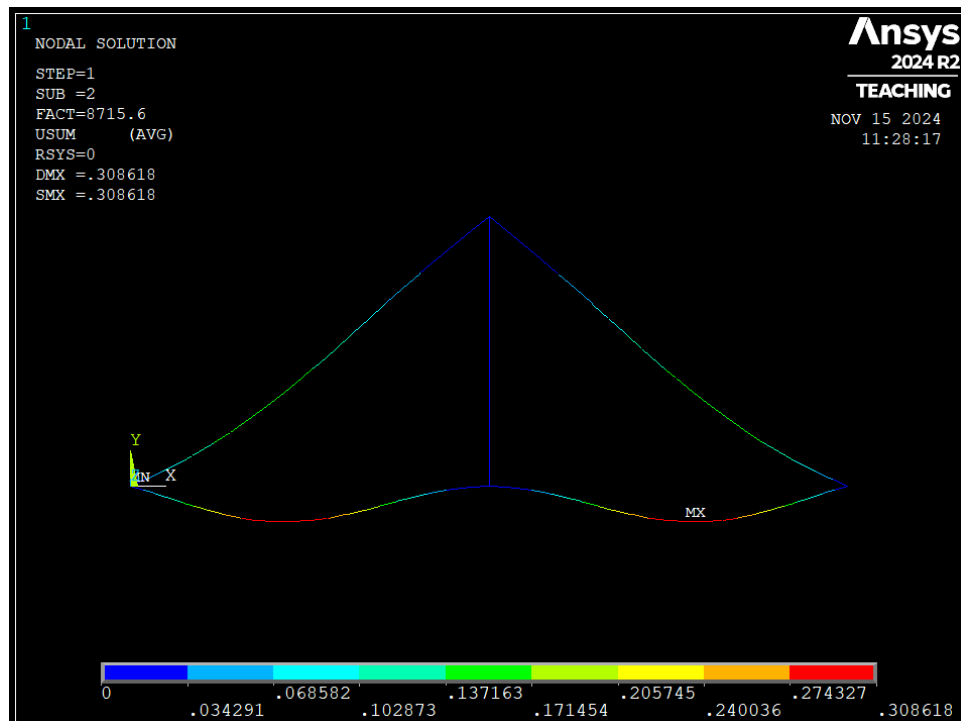
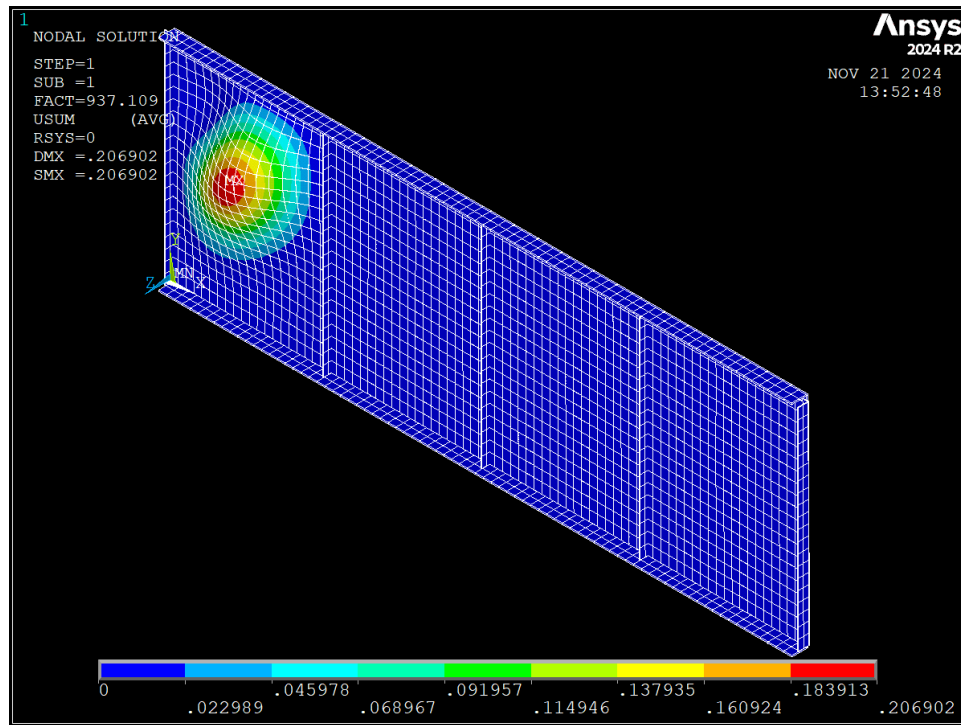
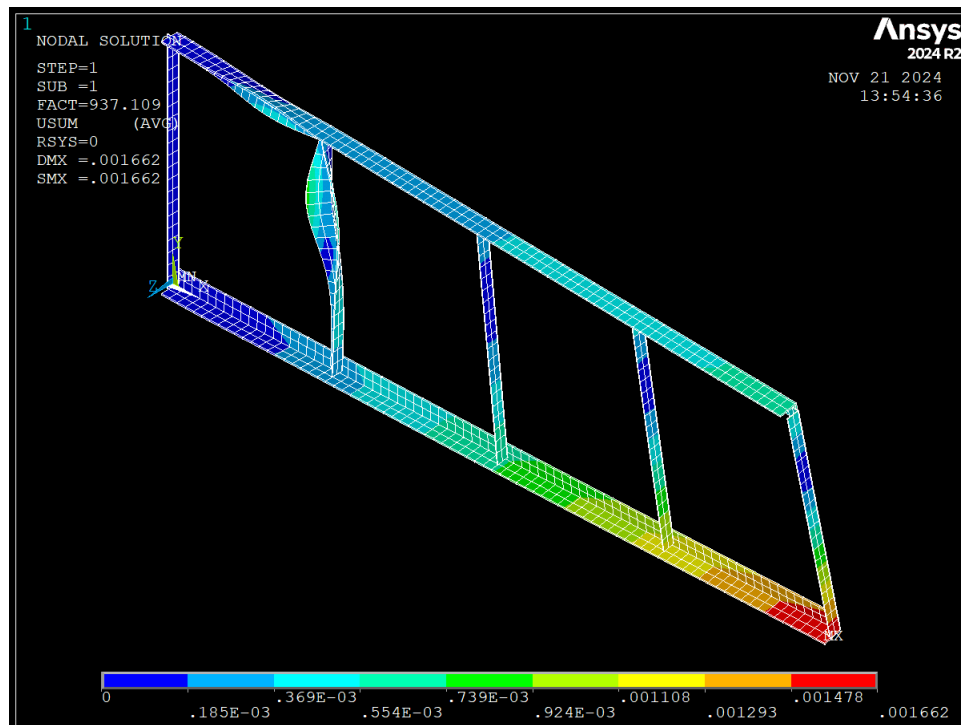


Figure 9. Problem 2 – Mode 2 Displacement

Part 3**Figure 10. Mode 1 Displacement – Entire Structure****Figure 11. Mode 1 Displacement – Stiffeners and Flanges Only**

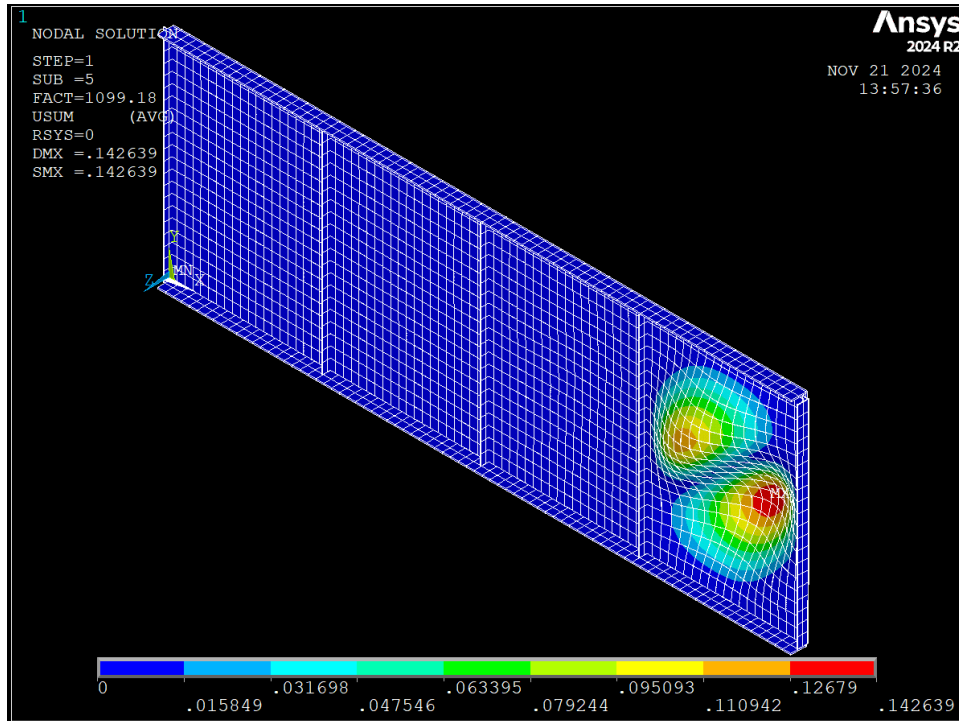


Figure 12. Mode 5 Displacement – Entire Structure

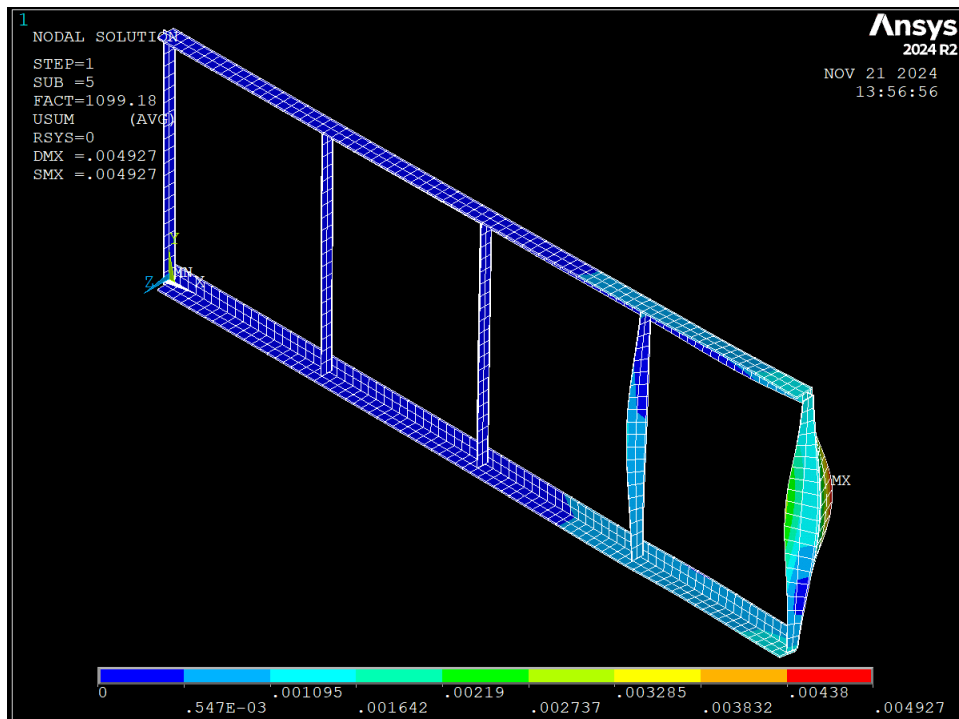


Figure 13. Mode 5 Displacement – Stiffeners and Flanges Only

Analysis

In Part 1, the beam is shown bending and buckling in a direction that is not aligned with the primary axes. This is due to the fact that buckling occurs along a principal axis. For convention, the x and y axes of the cross section are the y and z axes in this setup, respectively. Shown here (and confirmed in the problem sets), the x' axis is the weakest so buckling occurs there. That means that the buckling plane exists along the y' axis. The figure below illustrates these principal axes to show how the bending occurs out of the plane of the principal axis.

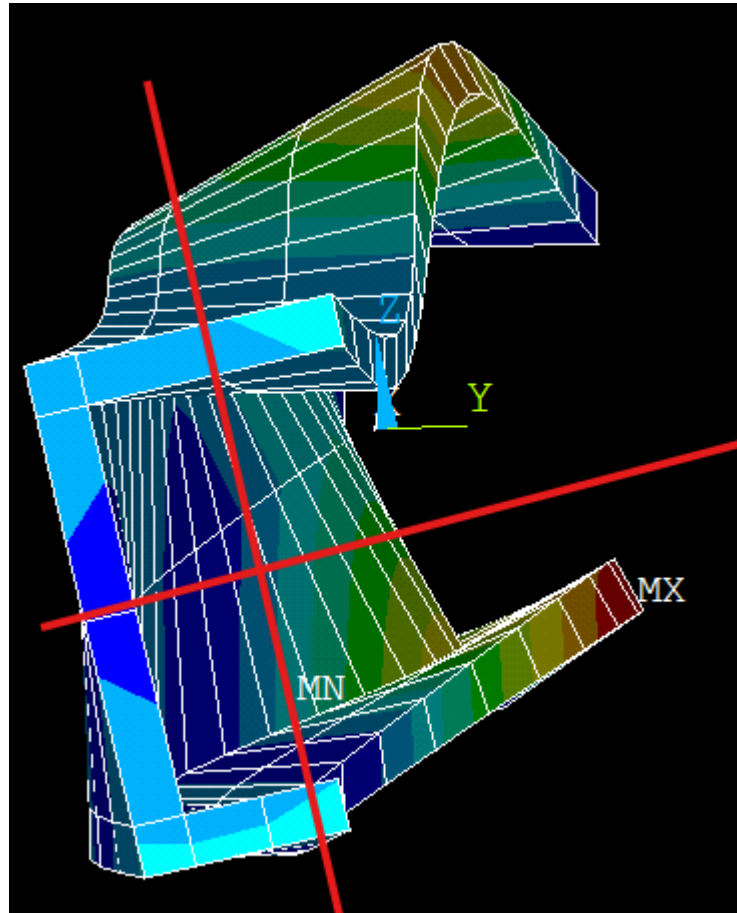


Figure 14. Principal axis

For Part 2, the calculated buckling loads were as follow:

	Buckling Load (N)
Problem 1 – Mode 1	5433
Problem 1 – Mode 2	7772
Problem 2 – Mode 1	5980
Problem 2 – Mode 2	6716

Conclusion

Overall, this lab was a comprehensive exercise in several different buckling scenarios in order to better visualize how the structure responds to a critical load. It highlighted the tendency to buckle along principal axes, as well as buckling in more complex structures.

Appendix – Source Codes

Part 1

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SECOFFSET, CENT
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k,2,0.5,0,0
l,1,2
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ETDEL,1
ETDEL,2
ET,1,BEAM188
LMESH, 1
LCLEAR, 1
SMRT,6
SMRT,7
SMRT,6
SMRT,OFF
FLST,5,1,4,ORDE,1
FITEM,5,1
CM,_Y,LINE
LSEL, , , ,P51X
CM,_Y1,LINE
CMSEL,,_Y
LESIZE,_Y1, , ,20, , , ,1
/UI,MESH,OFF
FLST,2,1,3,ORDE,1
FITEM,2,2
/GO
DK,P51X,UZ, , ,0,UY, , , , ,
FLST,2,1,3,ORDE,1
FITEM,2,1
/GO
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```
DK,P51X, , , ,0,ALL, , , , ,
FINISH
/SOL
ANTYPE,0
PSTRES,1
solve
FINISH
/PREP7
ALLSEL,ALL
FINISH
/SOL
solve
FINISH
/PREP7
LMESH,    1
FINISH
/SOL
solve
FINISH
/SOLUTION
ANTYPE,1
FINISH
/POST1
SET,LIST
SET,LAST
SET,LIST,2
FINISH
/SOL
ANTYPE,1
BUCOPT,LANB,1,0,0,RANGE
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MXPAND,1,1,500000,1,0.001,
solve
FINISH
/POST1
SET,FIRST
FINISH
/SOL
ANTYPE,0
OUTRES,ERASE
OUTRES,ALL,ALL
PSTRES,1
solve
FINISH
```

```
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MXPAND,1,1,500000,1,0.001,
solve
FINISH
/POST1
SET,FIRST
SET,LAST
FINISH
/SOL
FINISH
/POST1
SET,FIRST
FINISH
/SOL
ANTYPE,0
OUTRES,ERASE
OUTRES,ALL,ALL
PSTRES,1
solve
FINISH
/POST1
SET,FIRST
FINISH
/SOL
ANTYPE,1
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solve
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/POST1
SET,FIRST
/CFORMAT,32,0
FINISH
/SOL
ANTYPE,0
OUTRES,ERASE
OUTRES,ALL,ALL
PSTRES,1
solve
FINISH
/POST1
FINISH
```

```
/SOL
ANTYPE,0
solve
FINISH
/POST1
FINISH
/SOL
ANTYPE,0
FINISH
/PREP7
FLST,2,1,3,ORDE,1
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FKDELE,ALL,ALL
FLST,2,1,3,ORDE,1
FITEM,2,2
/GO
FK,P51X,FX,-1
FINISH
/SOL
ANTYPE,0
OUTRES,ERASE
OUTRES,ALL,ALL
solve
FINISH
/POST1
FINISH
/SOL
ANTYPE,1
BUCOPT,LANB,1,0,0,RANGE
MXPAND,1,1,500000,1,0.001,
solve
FINISH
/PREP7
FINISH
/POST1
SET,FIRST
SET,NEXT
SET,LAST
WPSTYLE,,,,,,,,0
/SOLU
```

```
ANTYPE,0
OUTRES,ERASE
OUTRES,ALL,ALL
PSTRES,1
solve
FINISH
/POST1
FINISH
/SOL
ANTYPE,1
BUCOPT,LANB,2,0,0,RANGE
MXPAND,2,1,500000,1,0.001,
solve
FINISH
/POST1
SET,FIRST
SET,NEXT
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Part 2

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SECOFFSET, CENT
SECDATA,0.008,0.01,0,0,0,0,0,0,0,0,0
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K,2,2,0,0,
K,3,1,0.75,0,
K,4,1,0,0,
LSTR, 1, 4
LSTR, 4, 2
LSTR, 2, 3
LSTR, 3, 1
LSTR, 3, 4
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FLST,5,3,4,ORDE,2
LMESH, ALL
FLST,2,1,3,ORDE,1
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FLST,2,1,3,ORDE,1
FITEM,2,2
/GO
DK,P51X,UZ,,0,UY,,,,,
FLST,2,2,3,ORDE,2
FITEM,2,3
FITEM,2,-4
/GO
DK,P51X,,,0,UZ,,,,,
FLST,2,4,4,ORDE,2
FITEM,2,1
FITEM,2,-4
/GO
DL,P51X,,UZ,
FLST,2,1,3,ORDE,1
FITEM,2,3
FINISH
/POST1
FINISH
/PREP7
FLST,2,1,3,ORDE,1
FITEM,2,2
/GO
FK,P51X,FX,-1
FINISH
/SOL
ANTYPE,0
PSTRES,1
SOLVE
FINISH
/SOLUTION
ANTYPE,1
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MXPAND,2,1,1000000,1,0.001,
SOLVE
```

Part 3

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FITEM,2,8
FITEM,2,9
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FLST,2,4,4
FITEM,2,2
FITEM,2,11
FITEM,2,7
FITEM,2,10
AL,P51X
FLST,2,4,4
FITEM,2,3
FITEM,2,12
FITEM,2,6
FITEM,2,11
AL,P51X
FLST,2,4,4
FITEM,2,4
FITEM,2,13
FITEM,2,5
FITEM,2,12
AL,P51X
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ET,2,BEAM188
ET,3,SHELL181
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SECOFFSET, CENT
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SECTYPE, 2, BEAM, L, stiff, 0
SECOFFSET, CENT
SECDATA,0.051,0.051,0.0032,0.0032,0,0,0,0,0,0,0
sect,3,shell,,skin
secdata, 0.001,1,0.0,3
secoffset,MID
seccontrol,,,, , , ,
K,1,0,0,0,
K,2,0.75,0,0,
K,3,1.5,0,0,
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 K,5,3,0,0,
 K,6,3,1,0,
 K,7,2.25,1,0,
 K,8,1.5,1,0,
 K,9,0.75,1,0,
 K,10,0,1,0,
 LSTR, 1, 2
 LSTR, 2, 3
 LSTR, 3, 4
 LSTR, 4, 5
 LSTR, 6, 7
 LSTR, 7, 8
 LSTR, 8, 9
 LSTR, 9, 10
 LSTR, 1, 10
 LSTR, 2, 9
 LSTR, 3, 8
 LSTR, 4, 7
 LSTR, 5, 6
 FLST,2,4,4
 FITEM,2,1
 FITEM,2,10
 FITEM,2,8
 FITEM,2,9
 AL,P51X
 FLST,2,4,4
 FITEM,2,2
 FITEM,2,11
 FITEM,2,7
 FITEM,2,10
 AL,P51X
 FLST,2,4,4
 FITEM,2,3
 FITEM,2,12
 FITEM,2,6
 FITEM,2,11
 AL,P51X
 FLST,2,4,4
 FITEM,2,4
 FITEM,2,13
 FITEM,2,5
 FITEM,2,12
 AL,P51X

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FITEM,5,-4
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ASEL,, , ,P51X
CM,_Y1,AREA
CMSEL,S,_Y
CMSEL,S,_Y1
AATT,1,,3,0,3
CMSEL,S,_Y
CMDELE,_Y
CMDELE,_Y1
FLST,5,8,4,ORDE,2
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FITEM,5,-8
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LSEL,, , ,P51X
CM,_Y1,LINE
CMSEL,S,_Y
CMSEL,S,_Y1
LATT,1,1,, , ,1
CMSEL,S,_Y
CMDELE,_Y
CMDELE,_Y1
FLST,5,5,4,ORDE,2
FITEM,5,9
FITEM,5,-13
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LSEL,, , ,P51X
CM,_Y1,LINE
CMSEL,S,_Y
CMSEL,S,_Y1
LATT,1,2,, , ,2
CMSEL,S,_Y
CMDELE,_Y
CMDELE,_Y1
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LMESH,ALL
AMESH,ALL
FLST,2,1,4,ORDE,1
FITEM,2,9
/GO
DL,P51X,ALL,
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/SOL
ANTYPE,0
PSTRES,1
SOLVE
FINISH
/SOLUTION
ANTYPE,1
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MXPAND,10,1,1000000,1,0.001,
SOLVE
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